Type MS · Explain chamber wall thickness and heart valves by their function

Answer 1 of 6 · Question 7 part b · 5 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number				
7	(a)	pulmonary artery;		1
	(b)	• Y is thicker; • blood pumped into aorta; • takes/distributes blood to organs/tissues/whole body; • requires high pressure • thicker wall means more muscle for stronger contraction;	ORA for each point	5
	(c)	any three of • heart muscle deprived of oxygen; • insufficient energy for muscle contraction/less aerobic respiration/(more) anaerobic respiration; • heart stops pumping blood/pumps less blood; • heart attack;		3
	(d)	• transfers CO_2 (from muscle cells) to blood capillary; • transfers oxygen from blood/capillary (to muscle cells); • transfers glucose from blood/capillary (to muscle cells);		3

Total 12 marks

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 1 of 6 · Question 7 part b · 5 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

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Type MS · Explain chamber wall thickness and heart valves by their function

Answer 2 of 6 · Question 1 part b.i · 3 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
1	(a)	(i)	B; (pulmonary artery) *A it is not the aorta* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(ii)	A; (aorta) *B it is not the pulmonary artery* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(iii)	in at top left and out through vessel; in at top right and out through vessel;		2
		(iv)	valves; prevent blood flowing other way/backflow;		2
	(b)	(i)	chamber width in cm P 1.0; Q 0.2; R 0.2; S 1.5;	1 correct = 1 mark 2 correct = 2 marks 3/4 correct = 3 marks	3
		(ii)	An answer that makes reference to three of the following: • The width corresponds to the amount of muscle; • S pumps blood furthest distance/whole body; • P pumps blood to lungs/short distance; • Q/R only receives blood/doesn't have to pump it very far/to ventricles;		3

Total : 12 marks

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 2 of 6 · Question 1 part b.i · 3 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 3 of 6 · Question 1 part b.ii · 3 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Question number					
1	(a)	(i)	B; (pulmonary artery) *A it is not the aorta* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(ii)	A; (aorta) *B it is not the pulmonary artery* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(iii)	in at top left and out through vessel; in at top right and out through vessel;		2
		(iv)	valves; prevent blood flowing other way/backflow;		2
	(b)	(i)	chamber width in cm P 1.0; Q 0.2; R 0.2; S 1.5;	1 correct = 1 mark 2 correct = 2 marks 3/4 correct = 3 marks	3
		(ii)	An answer that makes reference to three of the following: • The width corresponds to the amount of muscle; • S pumps blood furthest distance/whole body; • P pumps blood to lungs/short distance; • Q/R only receives blood/doesn't have to pump it very far/to ventricles;		3

Total : 12 marks

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 3 of 6 · Question 1 part b.ii · 3 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 4 of 6 · Question 5 part d · 2 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

5 (a)	• ACE inhibitor prevents/inhibits action of enzyme; • prevents formation of angiotensin II; • less(angiotensin ii) bind with receptors; • Losartan blocks/binds to receptors; • angiotensin ii cannot bind to receptor;		4
(b)	• arteries/arterioles; • muscles in walls; • can contract/constrict; • narrow lumen; • smaller volume/space(for blood flow);		4
(c)	• in normal/healthy kidney protein not found in urine; • protein is a big molecule; • should not pass through the glomerulus/basement membrane;		2
(d)	• damage to glomerulus/basement membrane; • stronger/intense/eq; <u>contractions</u>; • (which causes) <u>muscle</u> to get thicker;	mp2 must be due to mp1	2
(e)	• detoxification occurs in the liver; • can't break down Losartan if impaired/accumulation /build-up in body/bloodstream/liver; • results in (further) liver damage/liver failure/liver diseae;		2 **Total =14**

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 4 of 6 · Question 5 part d · 2 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks
	• A=rods/cones/retina;		

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 5 of 6 · Question 1 part a.iv · 2 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

Question number					
1	(a)	(i)	B; (pulmonary artery) *A it is not the aorta* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(ii)	A; (aorta) *B it is not the pulmonary artery* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(iii)	in at top left and out through vessel; in at top right and out through vessel;		2
		(iv)	valves; prevent blood flowing other way/backflow;		2
	(b)	(i)	chamber width in cm P 1.0; Q 0.2; R 0.2; S 1.5;	1 correct = 1 mark 2 correct = 2 marks 3/4 correct = 3 marks	3
		(ii)	An answer that makes reference to three of the following: • The width corresponds to the amount of muscle; • S pumps blood furthest distance/whole body; • P pumps blood to lungs/short distance; • Q/R only receives blood/doesn't have to pump it very far/to ventricles;		3

Total : 12 marks

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 5 of 6 · Question 1 part a.iv · 2 marks

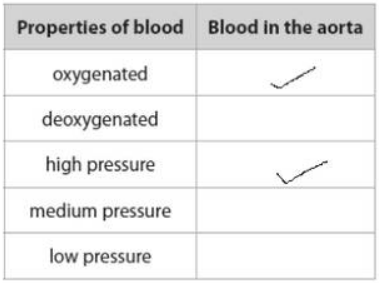
2022 JAN · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 6 of 6 · Question 2 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
2	(a)	(i)	left ventricle;		1
		(ii)	Two from: • valve; • prevents the backflow of blood; • so the blood flows the correct way around the heart/in one direction;	Ignore names of valves	2
		(iii)	 Properties of blood / Blood in the aorta oxygenated ✓ deoxygenated high pressure ✓ medium pressure low pressure	One mark removed for every additional incorrect tick.	2
		(iv)	the lungs/pulmonary vein;		1
	(b)		Two from: • it could be diseased/damaged/fatty deposits/blocked arteries; • the tissue/blood type doesn't match; • it has been removed from person for too long; • wrong size;	accept named diseases allow risk of rejection	2
	(c)	(i)	C platelets;		1
		(ii)	• fibrinogen is not converted to fibrin/fibrin not produced; • so factor XIII has no substrate/fibrin cannot form the clot/mesh not formed;		2

Total 11 marks

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 6 of 6 · Question 2 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks